

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 3

Dynamic Regression, Error Correction, and Exogeneity

Prof. Pedro Valls Pereira

`pedro.valls@fgv.br`

ADL Models · Error Correction Models · Engle–Granger Cointegration · Weak Exogeneity ·
Strong Exogeneity · Super-Exogeneity · Policy Invariance

Contents

1	The Autoregressive Distributed Lag Model	2
1.1	Long-run multiplier	2
2	From ADL to the Error Correction Model	2
3	Engle–Granger Single-Equation Cointegration Test	3
3.1	Two-step procedure	3
3.2	Super-consistency	3
4	The Engle–Hendry–Richard (1983) Exogeneity Framework	3
4.1	Setup	4
4.2	Weak exogeneity	4
4.3	Strong exogeneity	4
4.4	Super-exogeneity and policy invariance	4
5	Applied Example: UK Money Demand	5
6	Core References	5

1. The Autoregressive Distributed Lag Model

The $ADL(p, q)$ model provides the general framework for dynamic regression with a weakly exogenous conditioning variable X_t :

$$Y_t = \alpha_0 + \sum_{i=1}^p \alpha_i Y_{t-i} + \sum_{j=0}^q \beta_j X_{t-j} + \varepsilon_t, \quad \varepsilon_t \sim \text{iid}(0, \sigma^2).$$

This nests static regression ($p = q = 0$, only β_0) and pure $AR(p)$ as special cases. The residuals are white noise by construction if p and q are chosen adequately, ensuring valid inference.

1.1. Long-run multiplier

In the long run $Y_t = Y_{t-1} = Y^*$ and $X_t = X_{t-1} = X^*$, so the long-run solution of $ADL(1,1)$ is:

$$Y^* = \frac{\alpha_0}{1 - \alpha_1} + \underbrace{\frac{\beta_0 + \beta_1}{1 - \alpha_1}}_{\theta} X^* = \mu + \theta X^*.$$

θ is the **long-run multiplier**: the eventual effect of a permanent unit increase in X on Y . The **impact multiplier** β_0 captures the same-period response; intermediate multipliers track the path of adjustment.

The long-run relationship is well-defined only if the stability condition $|\alpha_1| < 1$ holds. If $\alpha_1 = 1$ (unit root in Y_t), the long-run mean is undefined unless Y_t and X_t are cointegrated.

2. From ADL to the Error Correction Model

Subtract Y_{t-1} from both sides of $ADL(1,1)$ and rearrange:

$$\Delta Y_t = \beta_0 \Delta X_t - (1 - \alpha_1) \underbrace{[Y_{t-1} - \theta X_{t-1} - \mu]}_{\text{error correction term (ECT)}} + \varepsilon_t.$$

The three components have clear economic interpretations:

- $\beta_0 \Delta X_t$: short-run (contemporaneous) response to changes in X .
- $(1 - \alpha_1)$: **speed of adjustment** — how quickly the system returns to equilibrium. Must satisfy $0 < 1 - \alpha_1 \leq 1$.

- $[Y_{t-1} - \theta X_{t-1} - \mu]$: **disequilibrium error** — last period's deviation from the long-run relationship. If $Y_{t-1} > \theta X_{t-1} + \mu$, the system is above equilibrium and ΔY_t is pulled downward.

Theorem 2.1 (Granger Representation Theorem, Engle & Granger, 1987). *If $Y_t \sim I(1)$ and $X_t \sim I(1)$ and they are cointegrated, they always have an ECM representation, and vice versa. Cointegration is equivalent to having an error correction representation.*

For the general $ADL(p, q)$ the ECM takes the form:

$$\Delta Y_t = \sum_{i=1}^{p-1} \tilde{\alpha}_i \Delta Y_{t-i} + \beta_0 \Delta X_t + \sum_{j=1}^{q-1} \tilde{\beta}_j \Delta X_{t-j} - \pi \underbrace{[Y_{t-1} - \theta X_{t-1}]}_{\text{ECT}} + \varepsilon_t,$$

where $\pi = -(1 - \alpha_1 - \dots - \alpha_p) < 0$ is the adjustment coefficient.

3. Engle–Granger Single-Equation Cointegration Test

3.1. Two-step procedure

Step 1. Estimate the long-run (cointegrating) regression by OLS:

$$Y_t = \mu + \theta X_t + \hat{u}_t.$$

Step 2. Apply an ADF-type test to the residuals $\hat{u}_t$:

$$\Delta \hat{u}_t = \pi \hat{u}_{t-1} + \sum_{j=1}^k \gamma_j \Delta \hat{u}_{t-j} + \eta_t.$$

$H_0 : \pi = 0$ (no cointegration) vs. $H_1 : \pi < 0$ (cointegration).

Because $\hat{u}_t$ is an estimated residual, the critical values are more negative than standard DF critical values. Use MacKinnon (1991) response-surface critical values, which depend on the number of variables and the sample size. Typically at 5%: ≈ -3.34 for two variables with a constant.

3.2. Super-consistency

OLS in a cointegrating regression is **super-consistent**: $\hat{\theta} \rightarrow \theta$ at rate T rather than the usual $\sqrt{T}$. However, the limiting distribution is non-normal, so standard OLS standard errors cannot be used for inference on θ . Use DOLS (dynamic OLS with leads and lags of ΔX_t) or FMOLS for valid asymptotic inference.

4. The Engle–Hendry–Richard (1983) Exogeneity Framework

4.1. Setup

Let $\mathbf{X}_t = (Y_t, Z_t)'$. The joint density factors as:

$$f(Y_t, Z_t | \mathbf{X}_{t-1}) = f(Y_t | Z_t, \mathbf{X}_{t-1}; \psi) \cdot f(Z_t | \mathbf{X}_{t-1}; \lambda).$$

The parameters of interest ψ appear only in the conditional model; λ governs the marginal model for Z_t .

4.2. Weak exogeneity

Definition 4.1. Z_t is *weakly exogenous* for ψ if:

- (i) ψ is identifiable from the conditional model $f(Y_t | Z_t, \mathbf{X}_{t-1}; \psi)$ alone.
- (ii) ψ and λ are variation-free (no cross-restrictions).

Implication: Conditioning on Z_t involves no loss of information about ψ . The conditional model can be estimated without modelling the marginal process for Z_t .

Weak exogeneity is a statement about the *parameters* of the model: it says that ψ can be recovered from the conditional model alone. Granger non-causality is a statement about the *distribution*: whether past Y helps predict future Z . Each can hold without the other.

4.3. Strong exogeneity

Definition 4.2. Z_t is *strongly exogenous* for ψ if it is weakly exogenous for ψ and Y_t does not Granger-cause Z_t . **Implication:** Valid for both estimation (uses only conditional model) and multi-step forecasting (future paths of Z_t can be specified independently of Y_t).

4.4. Super-exogeneity and policy invariance

Definition 4.3. Z_t is *super-exogenous* for ψ if it is weakly exogenous for ψ and the parameters ψ in the conditional model are invariant to changes in the process generating Z_t .

Super-exogeneity directly addresses the **Lucas critique** (1976): if Z_t is super-exogenous, then policy interventions that alter the distribution of Z_t leave the structural parameters ψ unchanged, validating the conditional model for policy analysis.

1. Detect structural breaks in the marginal model for Z_t (e.g., using IIS or Chow tests). Obtain break dummies D_t .
2. Add D_t to the conditional model $Y_t | Z_t$.
3. If D_t are *insignificant* in the conditional model $\Rightarrow$ parameters ψ are invariant $\Rightarrow$ support for super-exogeneity.

5. Applied Example: UK Money Demand

Hendry & Ericsson (1991) estimated an ADL model for UK velocity $v_t = \log(M1/P)_t$ conditional on the local authority interest rate $R_{la,t}$. The ECM takes the form:

$$\Delta v_t = \beta_0 \Delta R_{la,t} - \pi \underbrace{[v_{t-1} - \theta R_{la,t-1}]}_{\text{ECT}} + \text{dummies} + \varepsilon_t.$$

- The ECT is constructed from the long-run cointegrating relation estimated in step 1.
- $\hat{\pi} < 0$ and significant: the system corrects back to the long-run equilibrium.
- Super-exogeneity of $R_{la,t}$: when the interest rate regime changed in 1985, the velocity equation remained structurally stable — the break in the marginal model of $R_{la,t}$ did not spill into the conditional model.

6. Core References

- **Engle, R.F. & Granger, C.W.J.** (1987). Co-integration and error correction. *Econometrica*, 55, 251–276.
- **Engle, R.F., Hendry, D.F. & Richard, J.-F.** (1983). Exogeneity. *Econometrica*, 51, 277–304.
- **Hendry, D.F. & Doornik, J.A.** (2014). *Empirical Model Discovery and Theory Evaluation*. MIT Press.
- **Hendry, D.F. & Ericsson, N.R.** (1991). Modeling narrow money demand in the UK and US. *European Economic Review*, 35, 833–881.
- **Pesaran, M.H.** (2015). *Time Series and Panel Data Econometrics*. OUP. Chapter 21.
- **Dubé, J.-P., Hitsch, G.J., & Manchanda, P.** (2005). An empirical model of advertising dynamics. *QME*, 3, 107–144.
- **Dekimpe, M.G. & Hanssens, D.M.** (1995). The persistence of marketing effects on sales. *Marketing Science*, 14, 1–21.